distance is much smaller than thickness of wire → not possible

8 In Singapore, the power stations are situated near the coast because sea water is used to condense steam into water. The transmission of electric power over long distances from the power stations to homes would not be feasible without transformers.

A step-up transformer near the power station boosts the station's output root-mean-square (r.m.s.) voltage from 12.0 kV to 240 kV and a series of step-down transformers near the homes reduces the r.m.s. voltage to a final value of 240 V at the homes. The transmission voltage has a frequency of 50 Hz.

(a) For the transformer located near the power station, determine the ratio

$$\frac{N_p}{N_s} = \frac{\text{number of turns of primary coil}}{\text{number of turns of secondary coil}}.$$

[1]

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$
$$= \frac{12000}{240000}$$
$$= 0.05$$

(b) Explain why

(i) the voltage is stepped up near the power plant, [2]

The same power delivered at **higher transmission voltage** would result in a **lower transmission current**.
Lower current results in **lower power loss/dissipated in the power cables**.

(ii) an alternating voltage is used instead of direct voltage to transmit electrical energy. [1]

Alternating voltage can set up a **changing magnetic flux linkage** in a transformer which can be used to step the transmission voltage up or down.

(c) **(i)** Determine the r.m.s. current delivered to an electric kettle rated at 700 W when it is connected to a power socket in the house. [1]

Kettle power rating = P_{avg}
$P_{avg} = V_{rms} I_{rms}$
$I_{rms} = 700/240 = 2.92$ A

(ii) Calculate the peak output voltage to the electric kettle. [1]

$V_0 = \sqrt{2} \times V_{rms} = \sqrt{2} \times 240$
$= 339$ V

(d) On Fig. 8.1 below, sketch a labelled graph for the variation with time t of the thermal power P dissipated by the electric kettle.

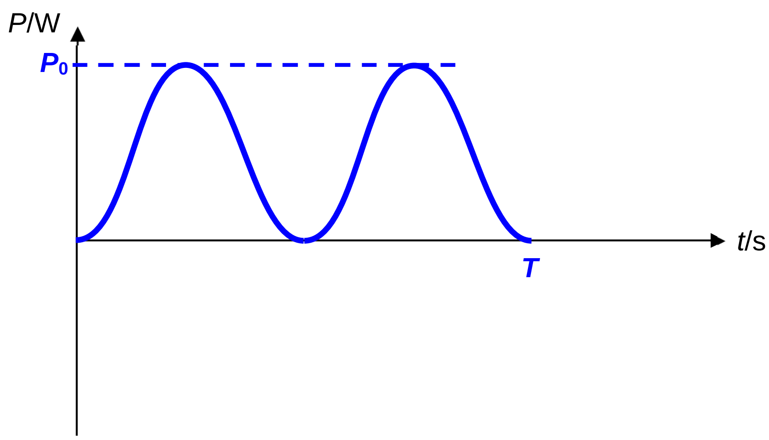

Fig. 8.1

[2]

9 The use of satellites for communication using microwaves is relatively commonplace, for

For